Functions

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UAB-IDEA

September 2026

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Section 1

Functions

What is a function?

Definition. Given nonempty sets X, Y, a **function** $f : X \to Y$ assigns to **every** $x \in X$ **exactly one** element $f(x) \in Y$.

1. every $x \in X$ is assigned some value in Y;
2. if $f(x) = y_1$ and $f(x) = y_2$, then $y_1 = y_2$.

X is the **domain**, Y the **codomain**. We typically take $X = \mathbb{R}$ for one variable or $X = \mathbb{R}^n$ for n variables.

Video: Introduction to Mathematical Thinking, *Function, domain, codomain, range*.

Range vs. codomain

Range. The image of the domain,

$$R(f) = f(X) = \{\, y \in Y : \exists x \in X,\ f(x) = y \,\}.$$

Codomain Y the target set.

Range $R(f)$ the set of values actually attained: a *subset* of Y.

Injective, surjective, bijective

Let $f : X \to Y$. These properties compare inputs with attained and outputs.

Injection (one-to-one)

$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.

Equivalently,

$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.

Surjection (onto)

$\forall y \in Y \; \exists x \in X$ with $f(x) = y$.

Equivalently, $f(X) = Y$.

Bijection

injective and surjective:

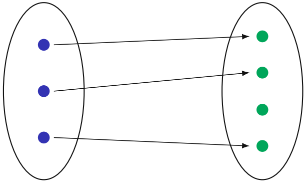
every output has exactly one input;

also called a *one-to-one correspondence*.

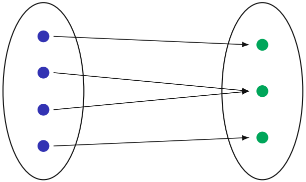
Further reading: Boyd, Sections 33.1–33.2.

Video: TrevTutor, *Injective, Surjective, and Bijective Functions*.

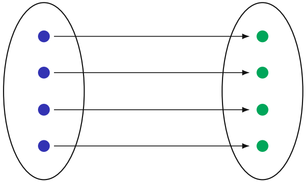
Three mapping patterns



Injection



Surjection



Bijection

Injectivity in econometrics

Identification of a parameter

A model attaches to each $\theta \in \Theta$ a distribution P_θ of the observable data. The parameter is identified when the map

$$\theta \mapsto P_\theta \quad \text{is injective:} \quad P_{\theta_1} = P_{\theta_2} \;\Rightarrow\; \theta_1 = \theta_2.$$

Two values sharing one P_θ are *observationally equivalent*: no sample size separates them.

BINARY CHOICE

With $y = \mathbf{1}\{x^T\beta + \varepsilon > 0\}$, $\varepsilon \mid x \sim N(0, \sigma^2)$ and Φ the standard normal CDF,

$$P(y = 1 \mid x) = \Phi\left(\frac{x^T\beta}{\sigma}\right).$$

WHERE INJECTIVITY FAILS

For every $c > 0$, $(c\beta, c\sigma)$ delivers the same choice probabilities, so θ and $c\theta$ share one P_θ; only β/σ enters those probabilities.

Normalizing $\sigma = 1$ removes this scale nonidentification. If $\beta \in \mathbb{R}^k$ and the support of x spans $\mathbb{R}^k$, strict monotonicity of Φ makes the normalized map injective.

Further reading: Lewbel, "The Identification Zoo," *Journal of Economic Literature* 57(4), pp. 835–903.

The graph of a function

Graph. For $f : X \to Y$,

$$\operatorname{graph}(f) = \{(x, f(x)) : x \in X\} \subseteq X \times Y.$$

For $f : X \subseteq \mathbb{R} \to \mathbb{R}$, equivalently,

$$\operatorname{graph}(f) = \{(x, y) \in X \times \mathbb{R} : y = f(x)\}.$$

Line tests on a graph

Vertical-line test A subset of $\mathbb{R}^2$ is the graph of a function of x exactly when each vertical line meets it in at most one point.

Horizontal-line test A function is injective exactly when each horizontal line meets its graph in at most one point.

Video: Mathispower4u, *Determining one-to-one and onto from a graph*.

Monotonicity and injectivity

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$.

Increasing	$x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$.
Strictly increasing	$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$.
Decreasing	$x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$.
Strictly decreasing	$x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$.

Strict monotonicity implies injectivity.

If $x_1 \neq x_2$, assume without loss of generality that $x_1 < x_2$. Then

$$\begin{aligned} f \text{ strictly increasing} \quad &\Rightarrow \quad f(x_1) < f(x_2), \\ f \text{ strictly decreasing} \quad &\Rightarrow \quad f(x_1) > f(x_2). \end{aligned}$$

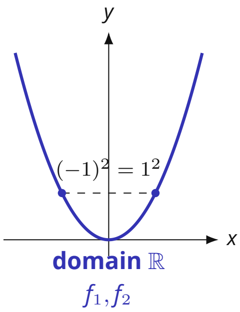
Hence $f(x_1) \neq f(x_2)$.

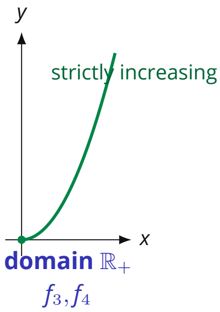
Same rule, different functions

All four functions use the same rule $x \mapsto x^2$. Only the domain and codomain change.

$$\begin{aligned} f_1 &: \mathbb{R} \to \mathbb{R}, & f_1(x) &= x^2, \\ f_2 &: \mathbb{R} \to \mathbb{R}_+, & f_2(x) &= x^2, \\ f_3 &: \mathbb{R}_+ \to \mathbb{R}, & f_3(x) &= x^2, \\ f_4 &: \mathbb{R}_+ \to \mathbb{R}_+, & f_4(x) &= x^2. \end{aligned}$$

For each f_i, decide: injection? surjection? bijection?





Restricting the domain changes the plotted graph. Changing only the codomain does not change the plotted points, but it can change whether the map is onto.

Same rule, different answers

	Injection	Surjection	Bijection
$f_1 : \mathbb{R} \to \mathbb{R}$	No	No	No
$f_2 : \mathbb{R} \to \mathbb{R}_+$	No	Yes	No
$f_3 : \mathbb{R}_+ \to \mathbb{R}$	Yes	No	No
$f_4 : \mathbb{R}_+ \to \mathbb{R}_+$	Yes	Yes	Yes

Why the answers differ

- On $\mathbb{R}$, $1^2 = (-1)^2$, so f_1 and f_2 are not injections.
- Into $\mathbb{R}$, no negative value is attained, so f_1 and f_3 are not surjections.
- On $\mathbb{R}_+$, $x \mapsto x^2$ is strictly increasing, so f_3 and f_4 are injections.
- With codomain $\mathbb{R}_+$, every $y \geq 0$ is attained by $x = \sqrt{y}$, so f_2 and f_4 are surjections.

The picture supports both conclusions: the full parabola fails the horizontal-line test, whereas the right-hand branch is injective. In all four cases the actual range is $\mathbb{R}_+$, so surjectivity holds exactly when the codomain is $\mathbb{R}_+$.

Restriction: changing the domain

Restriction. If $f : X \to Y$ and $A \subseteq X$, the **restriction** of f to A is $f|_A : A \to Y$, defined by $f|_A(x) = f(x)$.

1. Restrict the domain

For $q : \mathbb{R} \to \mathbb{R}$, $q(x) = x^2$, we have $q(1) = q(-1)$.
On $\mathbb{R}_+$,

$$q|_{\mathbb{R}_+} : \mathbb{R}_+ \to \mathbb{R}, \qquad q|_{\mathbb{R}_+}(x) = x^2.$$

This map is injective and has range $\mathbb{R}_+$.

2. Choose the attained codomain

Since the range is $\mathbb{R}_+$, define

$$\widetilde{q} : \mathbb{R}_+ \to \mathbb{R}_+, \qquad \widetilde{q}(x) = x^2.$$

Then $\widetilde{q}$ is bijective and

$$\widetilde{q}^{-1}(y) = \sqrt{y}.$$

Restricting the domain can restore injectivity. Declaring the attained range as codomain makes the restricted map surjective.

Video: Khan Academy, *Restricting the domain of a function to make it invertible*.

Invertibility and composition

Inverse. If $f : X \to Y$ is a bijection, there is a unique $f^{-1} : Y \to X$ satisfying

$$f(x) = y \Leftrightarrow f^{-1}(y) = x.$$

Composition. For $f : X \to Y$ and $g : Y \to Z$,

$$g \circ f : X \to Z, \qquad (g \circ f)(x) = g(f(x)).$$

With $\mathrm{id}_X(x) = x$,

$$f^{-1} \circ f = \mathrm{id}_X, \qquad f \circ f^{-1} = \mathrm{id}_Y\,.$$

The notation f^{-1} denotes an inverse function here; it does not denote $1/f$.

Video: Mario's Math Tutoring, *Inverse functions*.

Inverse and composition

Let

$$f:\mathbb{R}\to\mathbb{R},\quad f(x)=2x+3,\qquad g:\mathbb{R}\to\mathbb{R},\quad g(x)=x^2.$$

Find $g\circ f$, $f\circ g$, and f^{-1}. Does $g^{-1}:\mathbb{R}\to\mathbb{R}$ exist?

Inverse and composition

$$(g \circ f)(x) = (2x + 3)^2,$$

$$(f \circ g)(x) = 2x^2 + 3.$$

$$f^{-1}(y) = \frac{y - 3}{2}.$$

The inverse $g^{-1} : \mathbb{R} \to \mathbb{R}$ does not exist because g is not bijective.

Algebra of composition

Properties preserved under composition

A composition of injective maps is injective.

A composition of surjective maps is surjective.

A composition of bijective maps is bijective.

If f and g are bijections, then

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Further reading: Osborne, functions and inverse functions.

Selected algebraic rules and formulas

INEQUALITIES AND ABSOLUTE VALUES

$$a < b,\ c < 0 \ \Rightarrow\ ac > bc.$$

$$|x| \leq r \quad \Leftrightarrow \quad -r \leq x \leq r, \qquad r \geq 0.$$

POWERS AND LOGARITHMS

For $x, y > 0$ and $a, b \in \mathbb{R}$,

$$x^a x^b = x^{a+b}, \qquad (x^a)^b = x^{ab},$$

$$\log(xy) = \log x + \log y, \qquad \log(x^a) = a \log x.$$

Quadratic zeros and domain checks

For $q(x) = ax^2 + bx + c$ with $a \neq 0$, let $\Delta = b^2 - 4ac$. If $\Delta \geq 0$, the real zeros of q are

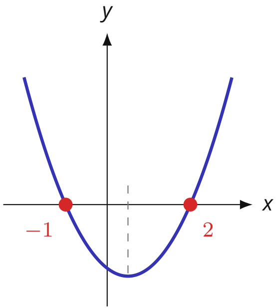
$$x = \frac{-b \pm \sqrt{\Delta}}{2a}.$$

If $\Delta < 0$, q has no real zero. When solving or simplifying equations, preserve the original domain restrictions; for example, logarithm arguments must be positive. After a non-reversible step such as squaring, verify the resulting values in the original equation.

Quick check. Solve $\log(x-1) + \log(x+1) = \log 8$, stating the domain before checking the algebraic solutions.

Answer. The domain is $x > 1$; $x^2 - 1 = 8$ gives $x = \pm 3$, so the solution is $x = 3$.

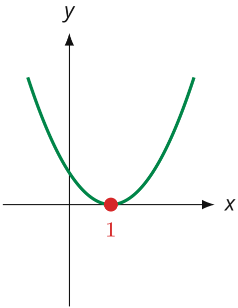
The sign of the discriminant on the graph



two crossings

$x^2 - x - 2$

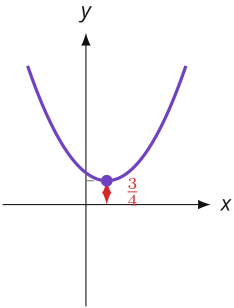
$\Delta = 9 > 0$



tangent to the axis

$x^2 - 2x + 1$

$\Delta = 0$



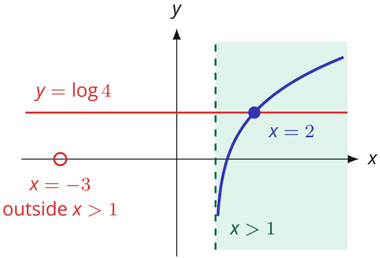
no crossing

$x^2 - x + 1$

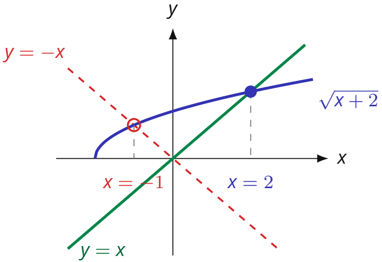
$\Delta = -3 < 0$

The same leading coefficient in all three panels: the vertex height $-\Delta/4a$ is what decides how often the graph meets the axis.

Two checks a candidate solution has to pass



$$\log(x-1) + \log(x+2) = \log 4$$



$$\sqrt{x+2} = x$$

Combining the two logarithms turns the equation into $x^2 + x - 6 = 0$, with roots 2 and -3. Only $x = 2$ lies in the shaded domain $x > 1$ on which both logarithms are defined.

Squaring turns the equation into $x^2 - x - 2 = 0$, with roots 2 and -1. At $x = -1$ the graph of $\sqrt{x+2}$ meets $y = -x$, so that root belongs to the sign-flipped equation.

Functions as indexed objects

A function is not only a formula such as $x \mapsto x^2$. Many familiar mathematical objects are functions once the index set is made explicit.

Indexed collections

An element $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ can be viewed as

$$x : \{1, \ldots, n\} \to \mathbb{R}, \qquad i \mapsto x_i.$$

A sequence $(x_k)_{k \geq 1}$ in X is a function

$$x : \mathbb{N} \to X, \qquad k \mapsto x_k.$$

Function space. For sets X, Y, write

$$Y^X = \{f : f : X \to Y\}$$

for the set of all functions from X to Y.

The exponent names the input set: each element of Y^X assigns one value in Y to every index in X.

Source: Mark Walker, Math Camp 2019, "Notation for Sets of Functions and Subsets," p. 1.

Indicator functions and the power set

Indicator functions

Let $S \subseteq X$. Its **indicator function** is

$$\mathbf{1}_S : X \to \{0,1\}, \qquad \mathbf{1}_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \notin S. \end{cases}$$

The subset can be recovered from the function:

$$S = \{\, x \in X : \mathbf{1}_S(x) = 1 \,\}.$$

Indicator functions and the power set

Subsets as functions

Subsets of X and functions $X \to \{0,1\}$ are in one-to-one correspondence. The set of all subsets of X, the **power set**, is therefore conventionally denoted

$$2^X.$$

If X and Y are finite, then $|Y^X| = |Y|^{|X|}$. In particular,

$$|2^X| = 2^{|X|}.$$

Further reading: Mark Walker, Math Camp 2019, “Notation for Sets of Functions and Subsets,” pp. 1–2.

Predicates as truth-valued functions

Quantifier order through a function

Represent an open statement $\phi(x, y)$ by

$$\phi : X \times Y \to \{T, F\}.$$

One choice works for every y

$$\exists x \in X \, \forall y \in Y : \phi(x, y) = T$$

The choice may depend on y

$$\forall y \in Y \, \exists x \in X : \phi(x, y) = T$$

The first statement implies the second because the same x can be used for every y. The reverse implication can fail when the selected x varies with y.

Function notation makes the dependence permitted by the quantifier order explicit.

Further reading: Mark Walker, Math Camp 2019, Exercise Set 1, Exercises 4–5.

Section 2

Images, preimages & level sets

Image and preimage

Let $f : X \to Y$.

Image. For $A \subseteq X$,

$$f(A) = \{f(x) : x \in A\} \subseteq Y.$$

In particular, $R(f) = f(X)$.

Preimage. For $B \subseteq Y$,

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \subseteq X.$$

The notation $f^{-1}(B)$ for a preimage does not require invertibility. If f is bijective, then $f^{-1}(\{y\}) = \{f^{-1}(y)\}$.

Video: Big Epsilon, *Images and preimages of sets under functions*.

Map properties from singleton preimages

Injectivity and surjectivity from preimages

For a single output $y \in Y$, the set $f^{-1}(\{y\})$ contains all inputs mapped to y. The earlier map properties are therefore characterized by singleton preimages:

$$\begin{aligned} f \text{ injective} \quad &\Leftrightarrow \quad |f^{-1}(\{y\})| \leq 1 \quad \forall y \in Y, \\ f \text{ surjective} \quad &\Leftrightarrow \quad |f^{-1}(\{y\})| \geq 1 \quad \forall y \in Y, \\ f \text{ bijective} \quad &\Leftrightarrow \quad |f^{-1}(\{y\})| = 1 \quad \forall y \in Y. \end{aligned}$$

Further reading: Osborne, functions and inverse functions.

Preimages and set operations

Preimages preserve set operations

Let $f : X \to Y$, let $(B_\alpha)_{\alpha \in A}$ be subsets of Y, and let $B \subseteq Y$. Then preimages preserve the basic set operations:

$$f^{-1}\left(\bigcup_{\alpha \in A} B_\alpha\right) = \bigcup_{\alpha \in A} f^{-1}(B_\alpha),$$

$$f^{-1}\left(\bigcap_{\alpha \in A} B_\alpha\right) = \bigcap_{\alpha \in A} f^{-1}(B_\alpha),$$

$$f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

Why preimages preserve structure

Pulling sets back to the domain

Statements about subsets of the codomain can be pulled back to statements about subsets of the domain. This is the basic set-theoretic mechanism behind measurable functions and the open-set characterization of continuity used later.

Further reading: MIT 18.S190, metric-space lecture notes.

Preimages in probability

Random variables

On a probability space $(\Omega, \mathcal{F}, P)$, a real-valued random variable is a measurable function $X : \Omega \to \mathbb{R}$. Events defined by Borel restrictions are preimages. For example,

$$\{X \leq t\} = X^{-1}((-\infty, t]) = \{\omega \in \Omega : X(\omega) \leq t\} \in \mathcal{F}.$$

Since $(-\infty, t]$ is a Borel set and X is measurable, its preimage belongs to $\mathcal{F}$ and is an event.

Exponential map

Exponential map.
$f\colon \mathbb{R}_+ \to \mathbb{R}_+$,

$$f(x) = e^x.$$

- The map is strictly increasing, so f is injective.
- It is not onto $\mathbb{R}_+$ because its range is $[1, \infty)$.
- $f([0,2]) = [1, e^2]$ and $f^{-1}((0,4]) = [0, \ln 4]$.

Decreasing affine map

Affine map.
$g\colon [0,\alpha/\beta] \to [0,\alpha]$,

$$g(x) = \alpha - \beta x, \qquad \alpha, \beta > 0.$$

- The map is strictly decreasing and bijective on this domain.
- Every value in $[0,\alpha]$ has one preimage.
- $g^{-1}(y) = (\alpha - y)/\beta$.

Another example

Define

$$f(x) = 10 - (x-2)^2, \qquad f : [0,4] \to [6,10].$$

- Surjective onto the codomain $[6, 10]$.
- Not injective: $f(1) = f(3) = 9$.
- Inputs equally far from 2 have the same image.
- For example, $f^{-1}(\{9\}) = \{1, 3\}$.

Definition

Level set. For $f : \mathbb{R}^2 \to \mathbb{R}$ and $c \in \mathbb{R}$,

$$L(f,c) = \{(x,y) \in \mathbb{R}^2 : f(x,y) = c\} = f^{-1}(\{c\}).$$

Thus a level set is the preimage of a singleton.

The same preimage language gives the sets used later in optimization:

$$\{f \leq c\} = f^{-1}((-\infty, c]), \qquad \{f \geq c\} = f^{-1}([c, \infty)).$$

Later connection: Boyd, Section 21.5 (upper and lower contour sets and quasiconcavity).

Video: MIT 18.02SC, *Level curves*.

$f(x, y) = x^2 + y^2$

- For $c > 0$, the level set $x^2 + y^2 = c$ is a circle of radius $\sqrt{c}$.
- For $c = 0$, the level set is $\{(0, 0)\}$; for $c < 0$, it is empty.
- Equal positive increments in c produce progressively smaller increases in $\sqrt{c}$, so the circles move closer together.

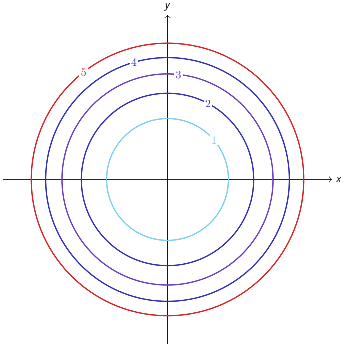


Figure 1 · Level curves at $c = 1, 2, 3, 4, 5$.

The paraboloid behind the rings

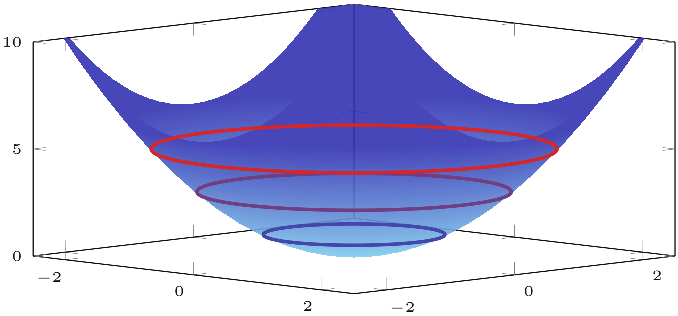


Figure 2 · $f(x,y) = x^2 + y^2$ *in* $3D$*, with its level rings at* $c = 1, 3, 5$.

Cobb-Douglas isoquants

Production isoquants

For $f : \mathbb{R}^2_{++} \to \mathbb{R}_{++}$ given by

$$f(x, y) = x^{0.7} y^{0.4},$$

a level $c > 0$ satisfies

$$f(x, y) = c \quad \Rightarrow \quad y = \left(\frac{c}{x^{0.7}}\right)^{1/0.4}.$$

In economics, level curves of production functions are called **isoquants**; level curves of utility functions are **indifference curves**.

Related later material: Boyd, Sections 20.1–20.2.

Images, preimages and level sets

1. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Find $f([-2, 1])$ and $f^{-1}([1, 4])$.
2. Let $h : \mathbb{R}^2 \to \mathbb{R}$, $h(x, y) = x + y$. Find and sketch $L(h, 2)$.
3. Let $u : \mathbb{R}^2_{++} \to \mathbb{R}_{++}$, $u(x, y) = xy$. For $c > 0$, describe $L(u, c)$ and write it as a preimage.

Images, preimages and level sets

Image and preimage

$$f([-2,1]) = [0,4],$$

$$f^{-1}([1,4]) = [-2,-1] \cup [1,2].$$

Linear level set

$$L(h,2) = \{(x,y) : x+y=2\} = \{(x,2-x) : x \in \mathbb{R}\}.$$

Cobb–Douglas level set

$$L(u,c) = u^{-1}(\{c\}) = \left\{\left(x,\frac{c}{x}\right) : x > 0\right\}.$$

It is the positive part of the hyperbola $xy = c$.

Section 3

Metrics, open & closed sets

Distances on $\mathbb{R}^n$

On $\mathbb{R}^n$, three frequently used metrics are:

$$d_1(x,y) = \sum_i |x_i - y_i|,$$

$$d_2(x,y) = \sqrt{\sum_i (x_i - y_i)^2},$$

$$d_\infty(x,y) = \max_i |x_i - y_i|.$$

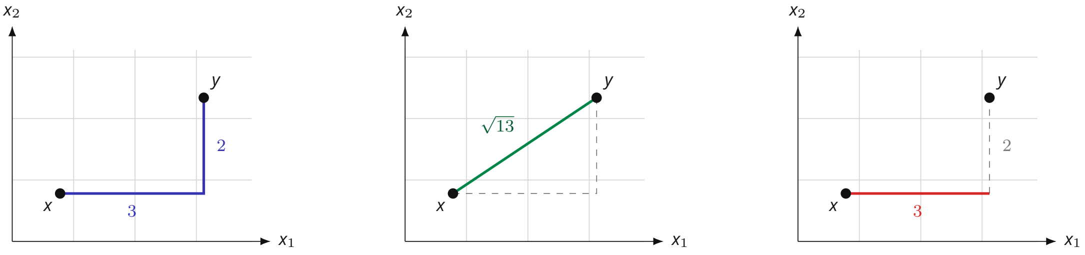
Metric axioms.
A function $d : X \times X \to \mathbb{R}$ is a metric if, for all $x, y, z \in X$,

1. $d(x,y) \geq 0$, with equality if and only if $x = y$;
2. $d(x,y) = d(y,x)$;
3. $d(x,z) \leq d(x,y) + d(y,z)$.

In this course, distance on $\mathbb{R}^n$ means the Euclidean metric d_2 unless stated otherwise.

Video: Bazett, *Weird notions of distance*. Further reading: Boyd, Ch. 10, §§10.22–10.49.

The three distances between one pair of points

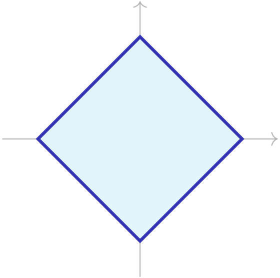


$d_1 = 3 + 2 = 5$ $\qquad$ $d_2 = \sqrt{3^2 + 2^2} = \sqrt{13}$ $\qquad$ $d_\infty = \max\{3, 2\} = 3$

Everywhere $x = (1,1)$ and $y = (4,3)$, so the coordinate gaps are 3 and 2: d_1 adds them, d_2 takes the hypotenuse they span, and d_∞ keeps the larger one.

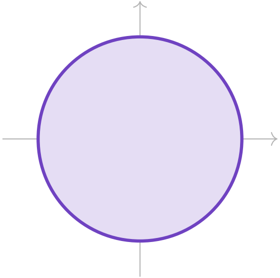
Unit balls for three metrics on $\mathbb{R}^2$

The closed unit ball is $\overline{B}^d_1(0) = \{x : d(x,0) \leq 1\}$. Its boundary depends on the metric.



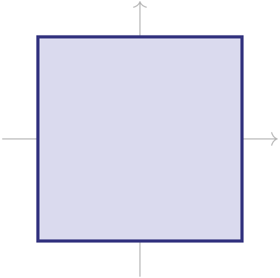
$|x_1| + |x_2| \leq 1$

d_1: diamond



$x_1^2 + x_2^2 \leq 1$

d_2: disc



$\max\{|x_1|, |x_2|\} \leq 1$

d_∞: square

Discrete metric

Discrete metric. For any set X, define

$$d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y, \end{cases} \qquad x,y \in X.$$

Triangle inequality for the discrete metric

Nonnegativity, nondegeneracy and symmetry are immediate. If $x = y$, then $d(x,y) = 0 \leq d(x,z) + d(z,y)$. If $x \neq y$, at least one of $x \neq z$ and $y \neq z$ holds, so $d(x,z)$ and $d(z,y)$ are not both zero; hence $1 \leq d(x,z) + d(z,y)$. ■

Further reading: MIT 18.S190, metric-space lecture notes.

The open ball

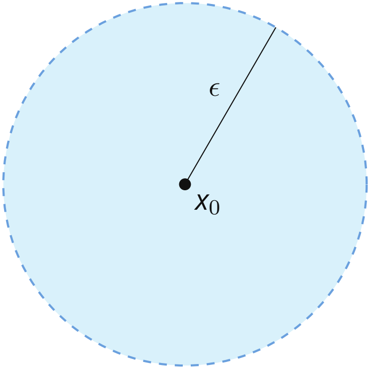
Open ball. In a metric space (X, d), the open ball of radius $\epsilon > 0$ around $x_0 \in X$ is

$$B_\epsilon^X(x_0) = \{ x \in X : d(x, x_0) < \epsilon \}.$$

If $X \subseteq \mathbb{R}^n$ inherits the Euclidean distance, then

$$B_\epsilon^X(x_0) = X \cap B_\epsilon^{\mathbb{R}^n}(x_0).$$

The ambient space matters: points outside X are not part of the ball.



$B_\epsilon^X(x_0)$ uses only points belonging to X.

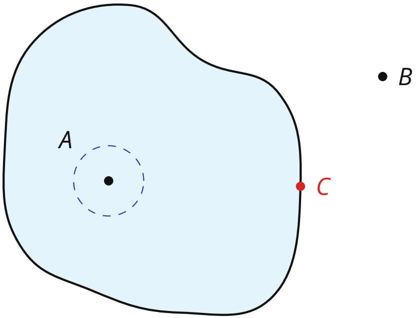
Video: NCX Maths, *Open ball, interior point, exterior point, boundary point*.

Interior & boundary points

Interior point. Let $S \subseteq X$. A point $s_0 \in S$ is an interior point of S in X if some $\epsilon > 0$ satisfies $B^X_\epsilon(s_0) \subseteq S$.

Boundary point. A point $x \in X$ is a boundary point of S in X if every $B^X_\epsilon(x)$ contains a point of S and a point of $X \setminus S$.

A is an interior point, *B* is an exterior point, and *C* is a boundary point.



Open, closed, closure

Open set. A set $S \subseteq X$ is open in X if every point of S is an interior point of S in X.

Closed set. A set $S \subseteq X$ is closed in X if its complement $X \setminus S$ is open in X.

Closure. The closure $\overline{S}$ in X is the union of S with all of its boundary points in X.

Example. In $X = (1,3] \cup (4,\infty)$, let $C = (1,2] \subseteq X$. Then $\partial_X C = \{2\}$ and $\overline{C}^X = C$. Since $1 \notin X$, the point 1 is not a boundary point of C in X.

Further reading: Boyd, Chapter 12, Sections 12.16–12.37.

Video: Wrath of Math, *Closed sets, closures, and clopen sets*.

The axioms behind "open"

Topology

A **topology** on a set X is a family $\tau \subseteq 2^X$ such that

1. $\varnothing \in \tau$ and $X \in \tau$;
2. every union of members of τ belongs to τ;
3. every intersection of finitely many members of τ belongs to τ.

The members of τ are the **open sets** and the pair (X, τ) is a **topological space**.

THE METRIC TOPOLOGY

On a metric space (X, d), take τ to be the unions of open balls $B_\epsilon^X(x)$. The axioms hold, and $S \in \tau$ exactly when every point of S is an interior point of S in X.

WHY FINITELY MANY

Each $\left(-\frac{1}{n}, \frac{1}{n}\right)$ is open in $\mathbb{R}$, while

$$\bigcap_{n \geq 1} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$$

is not open in $\mathbb{R}$. Hence arbitrary intersections of open sets need not be open.

Ambient space matters

An endpoint can belong to an open set.

Let

$$X = (1,3] \cup (4,\infty), \qquad d(x,y) = |x-y|, \qquad A = (1,3] \subseteq X.$$

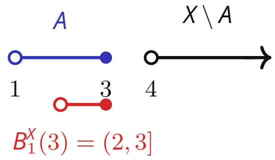
For $x \in (1,3)$, choose

$$\epsilon = \frac{1}{2}\min\{x-1,\ 3-x\} > 0.$$

Then $B^X_\epsilon(x) \subseteq A$. At the endpoint $x = 3$,

$$B^X_1(3) = X \cap (2,4) = (2,3] \subseteq A.$$

Hence every point of A is interior in X, so A is open in X.

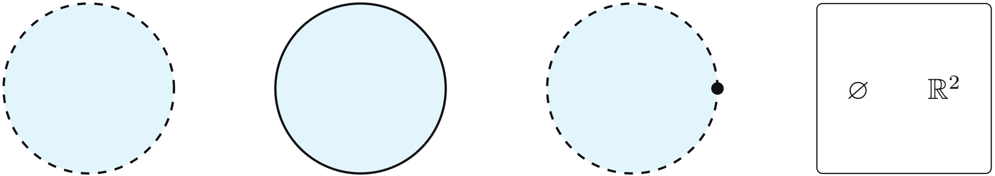


A is open in X.

A is not open in $\mathbb{R}$.

Also $X \setminus A = (4,\infty)$ is open in X, so A is both open and closed in X.

Open, closed, both, or neither



Open	**Closed**	**Neither**	**Both**
$B(0,1)$	$\overline{B}(0,1)$	$B(0,1) \cup \{(1,0)\}$	open and closed

Figure 4 · Examples in the usual topology on $\mathbb{R}^2$.

Dashed boundary points are excluded; solid boundary points are included.

Open and closed sets

1. In $X = \mathbb{R}$, classify each set as open, closed, both, or neither:

$$(0, 1), \qquad [0, 1], \qquad (0, 1], \qquad [0, \infty).$$

2. For $S = (0, 1] \subseteq \mathbb{R}$, find $\operatorname{int}_{\mathbb{R}} S$, $\partial_{\mathbb{R}} S$, and $\overline{S}^{\mathbb{R}}$.
3. Now take $X = [0, 2]$ and $A = [0, 1] \subseteq X$. Find $\operatorname{int}_X A$, $\partial_X A$, and $\overline{A}^X$, and classify A in X.

Open and closed sets

Classification in $\mathbb{R}$

$(0,1)$	open, not closed
$[0,1]$	closed, not open
$(0,1]$	neither
$[0,\infty)$	closed, not open

For $S=(0,1]$,

$$\operatorname{int}_{\mathbb{R}} S=(0,1), \quad \partial_{\mathbb{R}} S=\{0,1\}, \quad \overline{S}^{\mathbb{R}}=[0,1].$$

Relative to $X=[0,2]$

$$\operatorname{int}_X A=[0,1),$$

$$\partial_X A=\{1\}, \qquad \overline{A}^X=[0,1].$$

Thus A is closed and not open in X. The point 0 is interior relative to X, so it is not a boundary point of A in X.

Section 4

Limits & continuity

Sequences and subsequences

A sequence in $\mathbb{R}^n$ is a function from $\mathbb{N}$ to $\mathbb{R}^n$, written $(x_k)_{k\geq 1}$.

Convergence $x_k \to x$ if for every $\epsilon > 0$ there is K such that $k \geq K \Rightarrow \|x_k - x\| < \epsilon$.

Subsequence (x_{k_j}) is obtained by choosing indices $k_1 < k_2 < \cdots$.

Bounded (x_k) is bounded if $\|x_k\| \leq M$ for some $M < \infty$ and every k.

Basic facts in $\mathbb{R}^n$.

- Every convergent sequence is bounded.
- Every subsequence of a convergent sequence has the same limit.
- Every bounded sequence has a convergent subsequence (Bolzano–Weierstrass).

Quick check. For $x_k = (-1)^k$ and $y_k = (-1)^k k$, decide whether each sequence is bounded and convergent, and give a convergent subsequence when possible.

Further reading: Boyd, Chapter 12, Sections 12.1–12.14.

Sequences and subsequences

$x_k = (-1)^k$ The sequence is bounded and divergent. Its even and odd subsequences satisfy

$$x_{2j} = 1, \qquad x_{2j-1} = -1.$$

$y_k = (-1)^k k$ The sequence is unbounded and divergent, and it has no convergent subsequence.

Near, might not at

A limit is about values *near* the point, not necessarily the value *at* the point.
Define

$$f(x) = \begin{cases} \dfrac{x^2-1}{x-1}, & x \neq 1, \\ 7, & x = 1. \end{cases}$$

For $x \neq 1$,

$$\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1.$$

Hence

$$\lim_{x\to 1} f(x) = 2, \qquad f(1) = 7.$$

The limit exists even though the point value is different.

Two equivalent definitions of a limit

For $f : \mathbb{R}^n \to \mathbb{R}$, $c \in \mathbb{R}^n$, and $L \in \mathbb{R}$, the statement $\lim_{x\to c} f(x) = L$ has two equivalent definitions.

ϵ–δ. For every $\epsilon > 0$, there is $\delta > 0$ such that

$$0 < \|x - c\| < \delta \Rightarrow |f(x) - L| < \epsilon.$$

Sequential. For every sequence $(x_k) \subseteq \mathbb{R}^n \setminus \{c\}$,

$$x_k \to c \Rightarrow f(x_k) \to L.$$

The sequential criterion quantifies over *every* sequence approaching c. Two such sequences with different output limits disprove a proposed limit.

Video: Wrath of Math, *Connecting function limits and sequence limits*.

One-sided limits and sequences

Let $g(x) = |x|/x$ for $x \neq 0$. The sign of x determines the value of $g(x)$.

One-sided calculation

If $x < 0$, then $|x| = -x$, so

$$g(x) = \frac{-x}{x} = -1.$$

If $x > 0$, then $|x| = x$, so

$$g(x) = \frac{x}{x} = 1.$$

Therefore

$$\lim_{x \to 0^-} g(x) = -1, \qquad \lim_{x \to 0^+} g(x) = 1.$$

Sequential calculation

Take

$$x_k = \frac{1}{k}, \qquad y_k = -\frac{1}{k}.$$

Then $x_k \to 0$ and $y_k \to 0$, but

$$g(x_k) = 1, \qquad g(y_k) = -1.$$

The two approaching sequences have different output limits, so $\lim_{x \to 0} g(x)$ does not exist.

Limit laws and direct substitution

Whenever $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist as real numbers:

- $\lim cf = c \lim f$.
- $\lim(g \pm f) = \lim g \pm \lim f$.
- $\lim(gf) = (\lim g)(\lim f)$.
- $\lim \dfrac{g}{f} = \dfrac{\lim g}{\lim f}$ if $\lim f \neq 0$.
- $\lim[f(x)]^n = [\lim f(x)]^n$ for $n \in \mathbb{N}$.
- If $f(x) > 0$ near a and $\lim f = L > 0$, then $\lim \ln f = \ln L$.

Direct substitution. It applies when the expression is continuous at the point; for example,

$$\lim_{x\to 2}(x^2 + 3x - 1) = 2^2 + 3 \cdot 2 - 1 = 9.$$

Video: Professor Dave Explains, *Limits and limit laws in calculus*.

Computing a $0/0$ limit: factor and cancel

Direct substitution gives $0/0$, so simplify the expression for nearby points.

1. Start

$$\lim_{x\to 1} \frac{x^2-1}{x-1}$$

2. Factor and cancel

$$\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1, \qquad x \neq 1.$$

3. Take the limit

$$\lim_{x\to 1}(x+1) = 2.$$

Cancellation is used only for $x \neq 1$. That is enough because a limit depends on values arbitrarily close to 1, not on the value at 1.

Computing a $0/0$ limit: conjugate

Direct substitution in

$$\lim_{x \to 1} \frac{\sqrt{x+3}-2}{x-1}$$

gives $0/0$. For $x \neq 1$, multiply by the conjugate:

$$\begin{aligned}
\frac{\sqrt{x+3}-2}{x-1} &= \frac{\sqrt{x+3}-2}{x-1} \cdot \frac{\sqrt{x+3}+2}{\sqrt{x+3}+2} \\
&= \frac{x+3-4}{(x-1)(\sqrt{x+3}+2)} = \frac{x-1}{(x-1)(\sqrt{x+3}+2)} \\
&= \frac{1}{\sqrt{x+3}+2}.
\end{aligned}$$

Therefore

$$\lim_{x \to 1} \frac{\sqrt{x+3}-2}{x-1} = \frac{1}{\sqrt{1+3}+2} = \frac{1}{4}.$$

When a limit fails to exist

Negating the definition. For a given L, the statement $\lim_{x\to a} f(x) = L$ fails when

$$\exists\,\epsilon > 0 \;\; \forall\,\delta > 0 \;\; \exists\,x : \quad 0 < |x - a| < \delta \quad \text{and} \quad |f(x) - L| \geq \epsilon.$$

The limit at a fails to exist when this holds for every $L \in \mathbb{R}$.

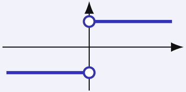
Sequential criterion. If $x_n \to a$ and $y_n \to a$ with $x_n, y_n \neq a$, and $f(x_n)$ and $f(y_n)$ converge to different values, then $\lim_{x\to a} f(x)$ fails to exist.

Three ways a limit can fail

DIFFERENT ONE-SIDED VALUES, UNBOUNDED BEHAVIOR, OSCILLATION

ONE-SIDED VALUES DIFFER

For $f(x) = |x|/x$ the limit from the left is -1 and from the right is 1.



UNBOUNDED BEHAVIOR

For $p(x) = 1/x$ the left limit is $-\infty$ and the right limit is $+\infty$, so the values leave every bounded interval as x nears 0.

OSCILLATION

For $r(x) = \sin(1/x)$, both $x_k = 1/(\pi/2 + 2\pi k)$ and $y_k = 1/(3\pi/2 + 2\pi k)$ tend to 0, while $r(x_k) = 1$ and $r(y_k) = -1$.

The sequential criterion settles the first and the third; the one-sided limits settle the second.

Video: Kimberly Brehm, *Limits that fail to exist*. Further reading: Hartman et al., *APEX Calculus* 3e, §1.4.

Several variables: different paths

In $\mathbb{R}^2$, the same point can be approached along many paths. Let

$$h(x,y) = \frac{2xy}{3x^2+y^2}.$$

PATH 1: $y=0$

$$h(x,0) = \frac{0}{3x^2} = 0.$$

PATH 2: $y=x$

$$h(x,x) = \frac{2x^2}{3x^2+x^2} = \frac{1}{2}.$$

The two path limits disagree: $0 \neq 1/2$. Therefore

$$\lim_{(x,y)\to(0,0)} \frac{2xy}{3x^2+y^2} \quad \text{does not exist.}$$

Video: Trefor Bazett, *Limits along paths*.

Several variables: proving existence

CONTROL THE ERROR UNIFORMLY IN ALL DIRECTIONS

Write $r = \|(x,y)-(a,b)\|$. A bound

$$|f(x,y)-L| \le h(r), \qquad h(r)\to 0 \text{ as } r\to 0,$$

depends on the distance alone, so one inequality covers every approach to (a,b) at once. A finite list of paths reaches the directions on the list; this bound reaches all of them.

Let

$$q(x,y) = \frac{x^2y}{x^2+y^2}.$$

Then, for $(x,y)\neq(0,0)$,

$$\left|\frac{x^2y}{x^2+y^2}\right| = |y|\,\frac{x^2}{x^2+y^2} \le |y| \le \sqrt{x^2+y^2} = r.$$

Since $r\to 0$, the squeeze theorem gives

$$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2} = 0.$$

Back to the first limit example

Recall the first limit example:

$$f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1, \\ 7, & x = 1. \end{cases}$$

For nearby points with $x \neq 1$, the formula simplifies to $f(x) = x + 1$. Hence

$$\lim_{x \to 1} f(x) = 2, \qquad f(1) = 7.$$

Continuity at 1 would require these two numbers to agree:

$$\lim_{x \to 1} f(x) = f(1).$$

Here $2 \neq 7$, so f is not continuous at 1.

If the point value were changed to $f(1) = 2$, the discontinuity would be removable.

Intuition & formalism

Intuition: inputs approaching x force outputs to approach $f(x)$.

Equivalently, in Euclidean spaces, continuity at x means

$$\lim_{x' \to x} f(x') = f(x).$$

For $f : \mathbb{R}^n \to \mathbb{R}^m$, f is **continuous at** x iff

$$\forall \epsilon > 0 \; \exists \delta > 0 \quad \|x' - x\| < \delta \Rightarrow \|f(x') - f(x)\| < \epsilon.$$

Continuity in metric spaces

In metric spaces (X, d) and (Y, d'), continuity at x means: for every $\epsilon > 0$ there exists $\delta > 0$ such that $d(x, x') < \delta \Rightarrow d'(f(x), f(x')) < \epsilon$. Equivalently, $x_n \xrightarrow{d} x \Rightarrow f(x_n) \xrightarrow{d'} f(x)$. See MIT 18.S190.

Video: Frank Swenton, *Continuity in metric spaces*.

A function on a finite domain can be continuous

A graph may look "broken" without violating continuity, because continuity is defined relative to the domain.

Consider

$$D = \{a, b\} \subseteq \mathbb{R}, \qquad a \neq b,$$

and any function

$$f : D \to \mathbb{R}.$$

For continuity at a, choose

$$\delta < \frac{|a-b|}{2}.$$

Then

$$x \in D,\ |x-a| < \delta \quad \Rightarrow \quad x = a,$$

so

$$|f(x) - f(a)| = 0 < \epsilon.$$

Therefore every function defined on a finite subset of $\mathbb{R}$ is continuous. The same argument applies at every point of the domain.

> Nearby points must belong to the domain. Missing points between a and b are irrelevant because they are not possible inputs.

Under the discrete metric every function is continuous

What made $D = \{a, b\}$ work is that each point sat alone in a small ball. The discrete metric arranges this on any set, finite or infinite.

Let X carry the discrete metric, let (Y, d') be any metric space, and let

$$f : X \to Y$$

be any function.

Given $\epsilon > 0$, take $\delta = 1$. Then

$$x' \in X,\ d(x', x) < 1 \quad \Rightarrow \quad x' = x,$$

so

$$d'(f(x'), f(x)) = 0 < \epsilon.$$

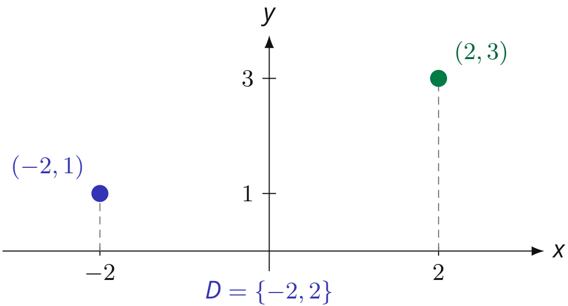
This δ works at every point of X, whatever ϵ was.

Any metric on a finite set does this too: if X has more than one point, $r = \min_{y \neq x} d(x, y) > 0$ gives $B_r^X(x) = \{x\}$; the singleton case is immediate. Thus a finite metric space has the discrete topology.

The graph of a function defined at two points

Let $D = \{-2, 2\}$ and define $f : D \to \mathbb{R}$ by

$$f(-2) = 1, \qquad f(2) = 3.$$



The graph is exactly

$$\text{graph}(f) = \{(-2, 1), (2, 3)\}.$$

No value of f is defined between the two plotted points.

Both points of D are isolated, so f is continuous at -2 and at 2.

Continuity via preimages

Open-set characterization

Let $f : X \to Y$ be a map between metric spaces. Then the following are equivalent:

1. f is continuous on X;
2. for every open set $U \subseteq Y$, the preimage $f^{-1}(U)$ is open in X.

Equivalently, $f^{-1}(C)$ is closed in X for every closed set $C \subseteq Y$.

This links the earlier notions directly:

$$\text{preimages} \;\to\; \text{open and closed sets} \;\to\; \text{continuity.}$$

Further reading: Boyd, Sections 13.12 and 13.29–13.31.

Properties

ALGEBRAIC CLOSURE

If $f, g : S \to \mathbb{R}$ are continuous, then so are

- $f + g$ and fg;
- f/g wherever $g \neq 0$.

COMPOSITION

If $f : X \to Y$ is continuous and $g : Y \to Z$ is continuous, then

$$g \circ f : X \to Z$$

is continuous.

Limits and continuity

1. Factor and cancel

$$\lim_{x\to 2} \frac{x^2-4}{x-2}.$$

2. Use a conjugate

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}.$$

3. One-sided behavior

$$\lim_{x\to 0} \frac{|x|}{x}.$$

Decide whether the limit exists and justify.

4. Different paths

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}.$$

Decide whether the limit exists and justify.

Limits and continuity

Factor and cancel

$$\frac{x^2-4}{x-2} = x+2 \quad (x \neq 2),$$
$$\lim_{x\to 2} \frac{x^2-4}{x-2} = 4.$$

Conjugate

$$\frac{\sqrt{1+x}-1}{x} = \frac{1}{\sqrt{1+x}+1} \quad (x \neq 0),$$

so the limit is $1/2$.

One-sided limits

$$\lim_{x\to 0^-} \frac{|x|}{x} = -1, \qquad \lim_{x\to 0^+} \frac{|x|}{x} = 1,$$

so the two-sided limit does not exist.

Path test Along $y=0$ the value is 0; along $y=x$ it is $1/2$. Hence the limit does not exist.

Section 5

Compactness & continuity theorems

Statement

Intermediate Value Theorem (IVT). If $f : [a,b] \to \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, then there exists $c \in [a,b]$ such that $f(c) = y$.

Equivalently, continuity forces the image to contain every value between the endpoint values:

$$[\min\{f(a), f(b)\}, \max\{f(a), f(b)\}] \subseteq f([a,b]).$$

- **Root-finding corollary.** If $f(a) > 0$ and $f(b) < 0$, then $0 \in f([a,b])$, so $f(c) = 0$ for some $c \in (a,b)$.
- This reconnects the IVT to the earlier image/range language: a continuous interval cannot skip an intermediate output value.

Further reading: Boyd, Section 29.60.

Video: Wrath of Math, *Intermediate Value Theorem and Finding Zeros*.

Every interior level is attained

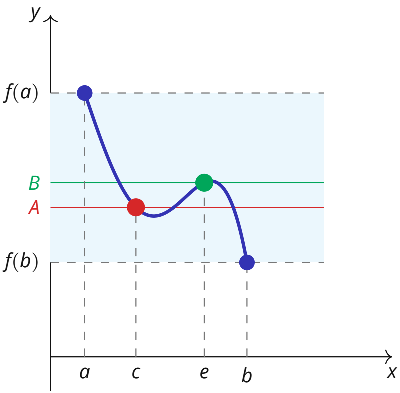


Figure 5 · Interior levels A and B.

Both levels lie between $f(a)$ and $f(b)$, and the continuous graph attains each one.

Intermediate value theorem

1. Use the Intermediate Value Theorem to show that

$$p(x) = x^7 - 5x^5 + x^3 - 1$$

   has a zero between -1 and 1.
2. Can the Intermediate Value Theorem be applied to $r(x) = 1/x$ on $[-1, 1]$ to show that r attains 0? State the assumption that fails.

Intermediate value theorem

Polynomial The function p is continuous, $p(-1) = 2 > 0$, and $p(1) = -4 < 0$. The IVT gives $c \in (-1, 1)$ with $p(c) = 0$.

Reciprocal function The function $r(x) = 1/x$ is not defined at 0, so it is not a continuous function on $[-1, 1]$. The IVT cannot be applied on that interval.

Closed and bounded in $\mathbb{R}^n$

Heine–Borel in $\mathbb{R}^n$. A set $K \subseteq \mathbb{R}^n$ is **compact** if and only if it is closed and bounded.

Sequence interpretation

Boundedness gives a convergent subsequence in $\mathbb{R}^n$; closedness keeps its limit in K. Hence every sequence in compact K has a subsequence converging to a point of K.

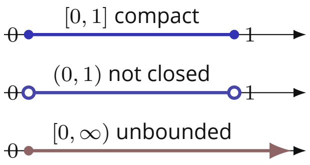


Figure 6 · Three subsets of $\mathbb{R}$.

In $\mathbb{R}^n$, compactness requires the set to be both closed and bounded.

More generally, the equivalence holds in every finite-dimensional normed space. In an infinite-dimensional normed space, compact sets are still closed and bounded, but the converse fails.

Video: Salomone, *Compact, closed and bounded*. Further reading: Boyd, §§29.41–29.46.

Beyond $\mathbb{R}^n$

Open covers and finite intersections

An open cover of K is a collection of open sets whose union contains K. A set $K \subseteq X$ is compact if every open cover of K has a finite subcover.

A collection $\{F_i\}_{i \in I}$ of closed sets has the **finite intersection property** relative to K when

$$K \cap F_{i_1} \cap \cdots \cap F_{i_n} \neq \varnothing \quad \text{for every finite subcollection.}$$

K is compact if and only if every such collection satisfies $K \cap \bigcap_{i \in I} F_i \neq \varnothing$.

Complements link the two: with $F_i = D_i^c$, an open cover with no finite subcover is a collection of closed sets with the finite intersection property whose total intersection misses K.

Existence arguments run on the closed sets. If K_1 is compact and $K_1 \supseteq K_2 \supseteq \cdots$ are nonempty closed subsets of K_1, every finite subcollection meets in its smallest member, so $\bigcap_{n \geq 1} K_n \neq \varnothing$.

Further reading: MIT 18.S190, Lecture 4, Definition 11 and Theorem 12; Munkres, *Topology*, Theorem 26.9; Rudin, *Principles of Mathematical Analysis*, Theorem 2.36.

Continuous functions on compact sets

Extreme Value Theorem (EVT). Let $K \subseteq \mathbb{R}^n$ be nonempty and compact. If $f : K \to \mathbb{R}$ is continuous, then f attains a maximum and a minimum on K.

This result is also often called the Weierstrass theorem.

The existence logic is exactly an image argument:

$$K \text{ compact} \overset{f \text{ continuous}}{\Rightarrow} f(K) \text{ compact in } \mathbb{R} \Rightarrow f(K) \text{ is closed and bounded.}$$

Hence $\sup f(K)$ and $\inf f(K)$ are finite, and closedness puts both values back inside $f(K)$. Therefore there exist $x^{\max}, x^{\min} \in K$ such that

$$f(x^{\min}) \leq f(x) \leq f(x^{\max}) \qquad \forall x \in K.$$

The theorem completes the chain: image $\to$ continuity $\to$ compact image $\to$ attained extrema.

Further reading: Boyd, Sections 30.1–30.2.

Video: Michael Penn, *The continuous image of a compact set*.

Semicontinuity on compact sets

One-sided continuity and existence

Let $K \subseteq \mathbb{R}^n$ and $f : K \to \mathbb{R}$. For every $a \in \mathbb{R}$,

$$\begin{aligned} f \text{ lower semicontinuous} \quad &\Leftrightarrow \quad \{x \in K : f(x) \le a\} \text{ is closed in } K, \\ f \text{ upper semicontinuous} \quad &\Leftrightarrow \quad \{x \in K : f(x) \ge a\} \text{ is closed in } K. \end{aligned}$$

On a nonempty compact set, lower semicontinuity guarantees a minimum and upper semicontinuity guarantees a maximum. A continuous function has both properties.

Further reading: Lafferriere, Lafferriere & Nguyen, Section 5.4, Theorems 5.4.3–5.4.5.

Existence in optimization

Existence of an optimizer

If a feasible set $K \subseteq \mathbb{R}^n$ is nonempty and compact, and the objective $u : K \to \mathbb{R}$ is continuous, then the problem

$$\max_{x \in K} u(x)$$

has a solution. This is the basic existence logic behind many consumer, producer and econometric optimization problems.

Further reading: Boyd, Section 30.3.

Compactness and the extreme value theorem

1. Decide which sets are compact in $\mathbb{R}$:

$$(0,1), \qquad [0,1], \qquad [0,\infty), \qquad K = \{1/n : n \in \mathbb{N}\} \cup \{0\}.$$

2. For each domain above, does the EVT guarantee that $f(x) = x$ attains a maximum? Does a maximum in fact exist?
3. Does $q(x) = 1/x$ attain a minimum on $(0,1]$? Which EVT assumption fails on this domain?

Compactness and the extreme value theorem

Compactness and $f(x) = x$

domain	compact?	maximum of f
$(0, 1)$	no	none
$[0, 1]$	yes	1
$[0, \infty)$	no	none
K	yes	1

The EVT guarantees the maximum on $[0, 1]$ and K.

Direct calculation on a noncompact domain The function $q(x) = 1/x$ is continuous on $(0, 1]$ and attains its minimum 1 at $x = 1$.

The domain $(0, 1]$ is not compact because it is not closed in $\mathbb{R}$. Hence the EVT does not supply this minimum; direct calculation does.

Log-likelihood replaces products by sums

The same maximizer in a simpler form

If observations are conditionally independent given their covariates, the likelihood is

$$L(\theta) = \prod_{i=1}^{N} f(y_i \mid x_i; \theta).$$

Suppose $L(\theta) > 0$ for every θ under consideration. Set $\ell(\theta) = \log L(\theta)$. Since the logarithm is strictly increasing,

$$\arg\max_{\theta} L(\theta) = \arg\max_{\theta} \ell(\theta), \qquad \ell(\theta) = \sum_{i=1}^{N} \log f(y_i \mid x_i; \theta).$$

The transformation preserves the maximizing parameter set and converts a product into a sum. Derivatives used for likelihood-based inference are therefore taken from ℓ.